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520

SOFTWARE CAT 3

MODULE 3,TERM 3

Instructions

Answer all questions

QUESTIONS

1. Define the term **API** and give one real-life example of its use. (2 marks)
2. State **three** differences between REST and SOAP. (3 marks)
3. Define an **API endpoint** and give one example. (2 marks)
4. Outline **four** main features of Postman. (4 marks)
5. State the HTTP method used to perform each of the following actions: (4 marks)
   (a) View student records
   (b) Add a new student
   (c) Edit a student's details
   (d) Remove a student record
6. State the meaning of each of the following HTTP status codes: (4 marks)
   (a) 201 (b) 401 (c) 404 (d) 500
7. Describe the steps you would follow in Postman to retrieve a list of users from https://jsonplaceholder.typicode.com/users using a GET request. (4 marks)
8. Define **corrective measures** and distinguish them from preventive and detective measures. (3 marks)
9. Name **four** software corrective tools and state the purpose of each. (4 marks)
10. Outline **five** steps followed when performing corrective measures on a faulty computer system. (5 marks)
11. Define **software testing** and state **two** of its objectives. (3 marks)
12. Differentiate between **smoke testing** and **regression testing**. (2 marks)
13. List **four** levels of software testing and briefly describe each. (4 marks)
14. Name the **four** types of software maintenance and give one example of each. (4 marks)
15. Define a **software maintenance schedule** and state **two** of its benefits. (2 marks)

CamScanner

1 API: A set of rules that allows different software applications to Communicate.
Example: A mobile banking app uses an API to retrieve account balances.

2 REST is an architectural style, while SOAP is a protocol.
REST Commonly uses JSON, while SOAP uses XML.
REST is generally lightweight, while SOAP has stricter messaging rules.

3 API endpoint → A specific URL where an API receives requests.
Example: https://jsonplaceholder.typicode.com/users

4 Sending API requests
Testing API responses
Automating API tests
Organizing requests into Collection.

5 View student records - GET
Add a new student - POST
Edit a students details - PUT or PATCH
Remove a student record - DELETE

6 201: Created Successfully
401: Unauthorized
404: Resource not found
500: Internal Server Error

7 Open Postman
Select the GET method.
Enter https://jsonplaceholder.typicode.com/users
Click Send and View the response.

8 Corrective: Fixes a fault after it occurs
Preventive: Stops faults from occurring
Detective: Identifies faults when they occur

9 Antivirus - Removes Malware
Debugger - Finds and fixes coding errors
System Restore - Restores the Computer to an earlier working state
Disk Repair tool - Detects and repairs disk errors

10 Identify the problem
Diagnose the cause
Back up important data
Apply the appropriate repair / fix
Test the system to confirm it works
document

11 Software testing is the process of Checking software to ensure it works correctly
Objectives
1) Identify errors and defects
2) Ensure the software meets users requirements

12 Smoke testing - Checks whether the main functions work after a new build
Regression testing - Checks whether changes have affected existing function

13 Unit testing - tests individual components
Integration testing - tests how components work together
System testing - tests the complete software system
Acceptance testing - Checks whether the software meets user requirements

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14. Corrective: Fixing a software bug
Adaptive - Updating software to work with a new operating system.
Perfective - Improving software performance
Preventive - Modifying code to reduce future errors 04

15 Software Maintenance Schedule - A plan showing when software maintenance activities should be performed 02

Benefits

a) Reduces unexpected system failures
b) Improves software performance and reliability - 06